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(54) **CUSHIONING STRUCTURE FOR
ANTI-COLLISION SPORTS PROTECTOR**

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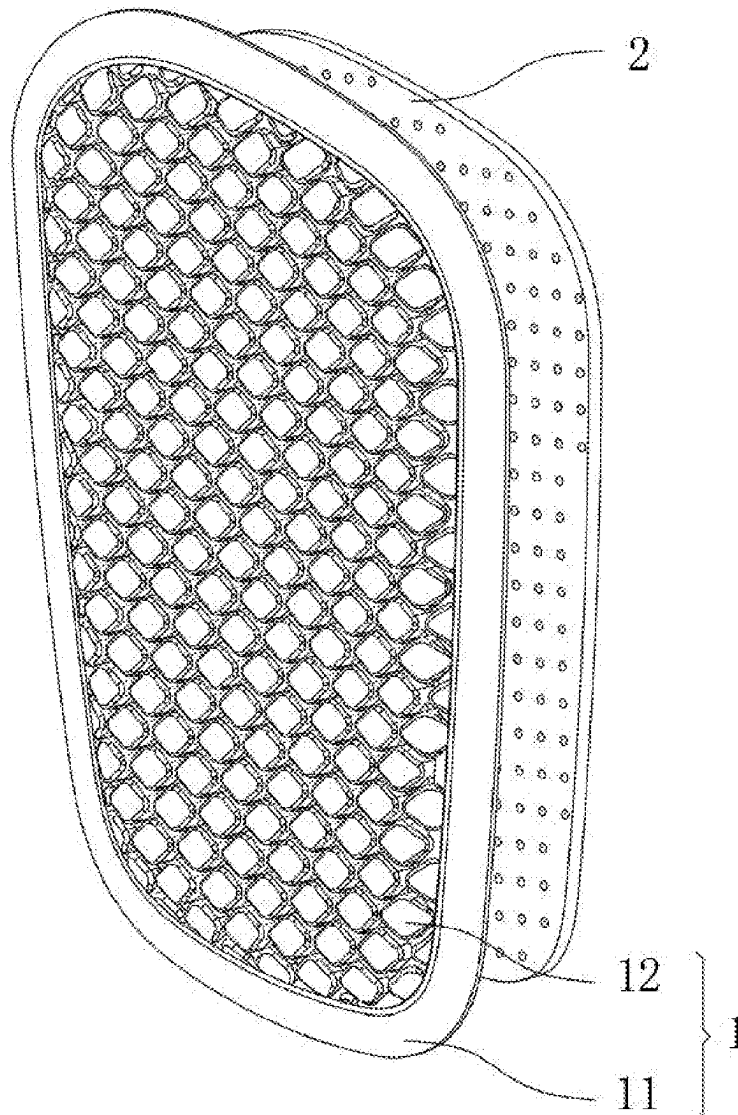
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(57) **ABSTRACT**

Disclosed is a cushioning structure for an anti-collision sports protector. The cushioning structure includes an energy-absorbing plate and a cushioning plate, and the energy-absorbing plate and the cushioning plate are both made of elastic materials. The energy-absorbing plate includes an annular frame, a plurality of energy-absorbing parts are arranged at an interval inside the annular frame, and two adjacent energy-absorbing parts are fixedly connected through a connecting block. A cavity is formed inside each of the energy-absorbing parts, a plurality of the energy-absorbing parts are combined to form an energy-absorbing net, an inner side of the annular frame is fixedly connected to the energy-absorbing net, and the cushioning plate covers an end of the annular frame. The present disclosure provides the cushioning structure for an anti-collision sports protector, which further reduces an impact force against the protector, and enhances comfort of using the protector.



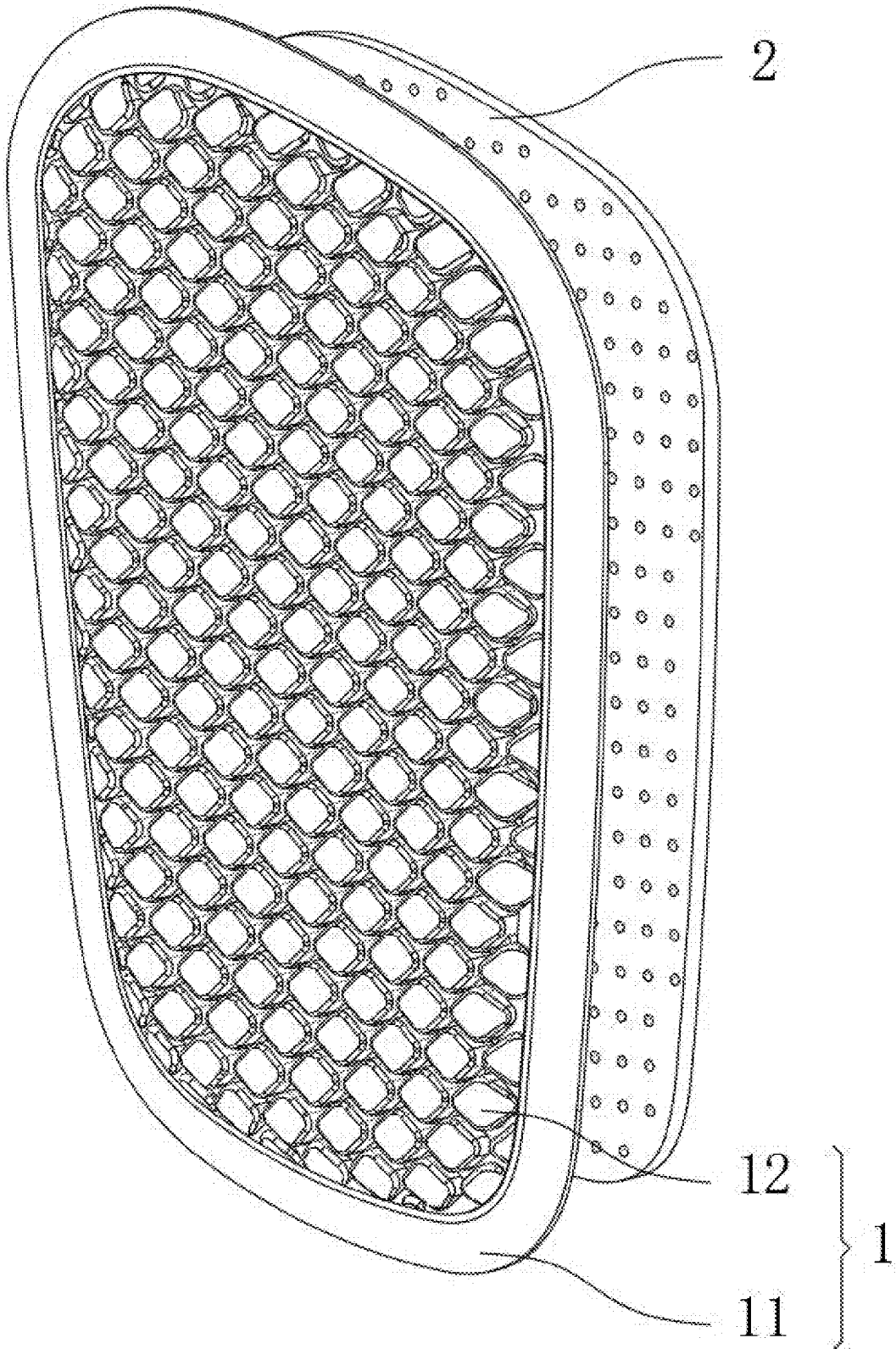


FIG. 1

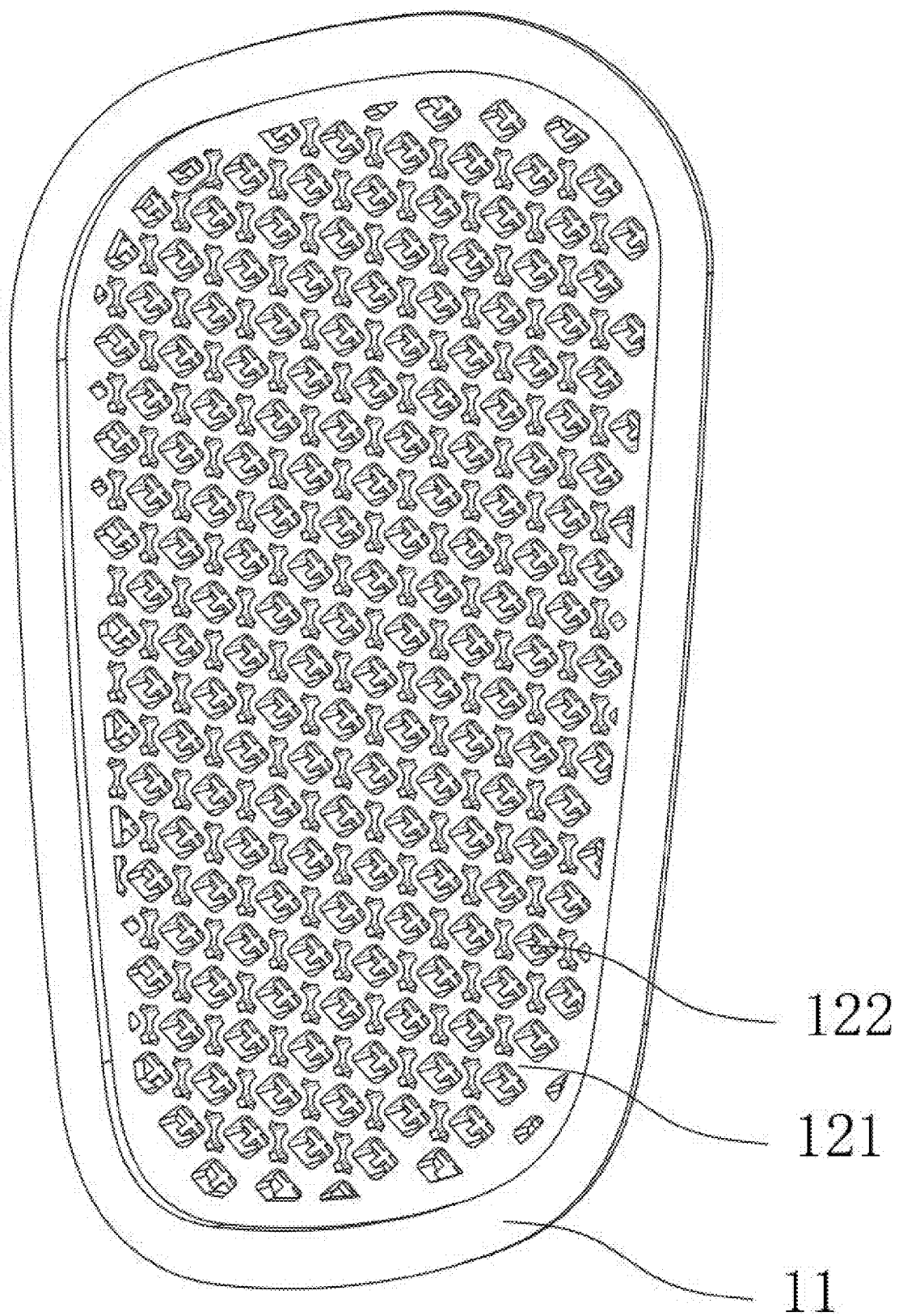


FIG. 2

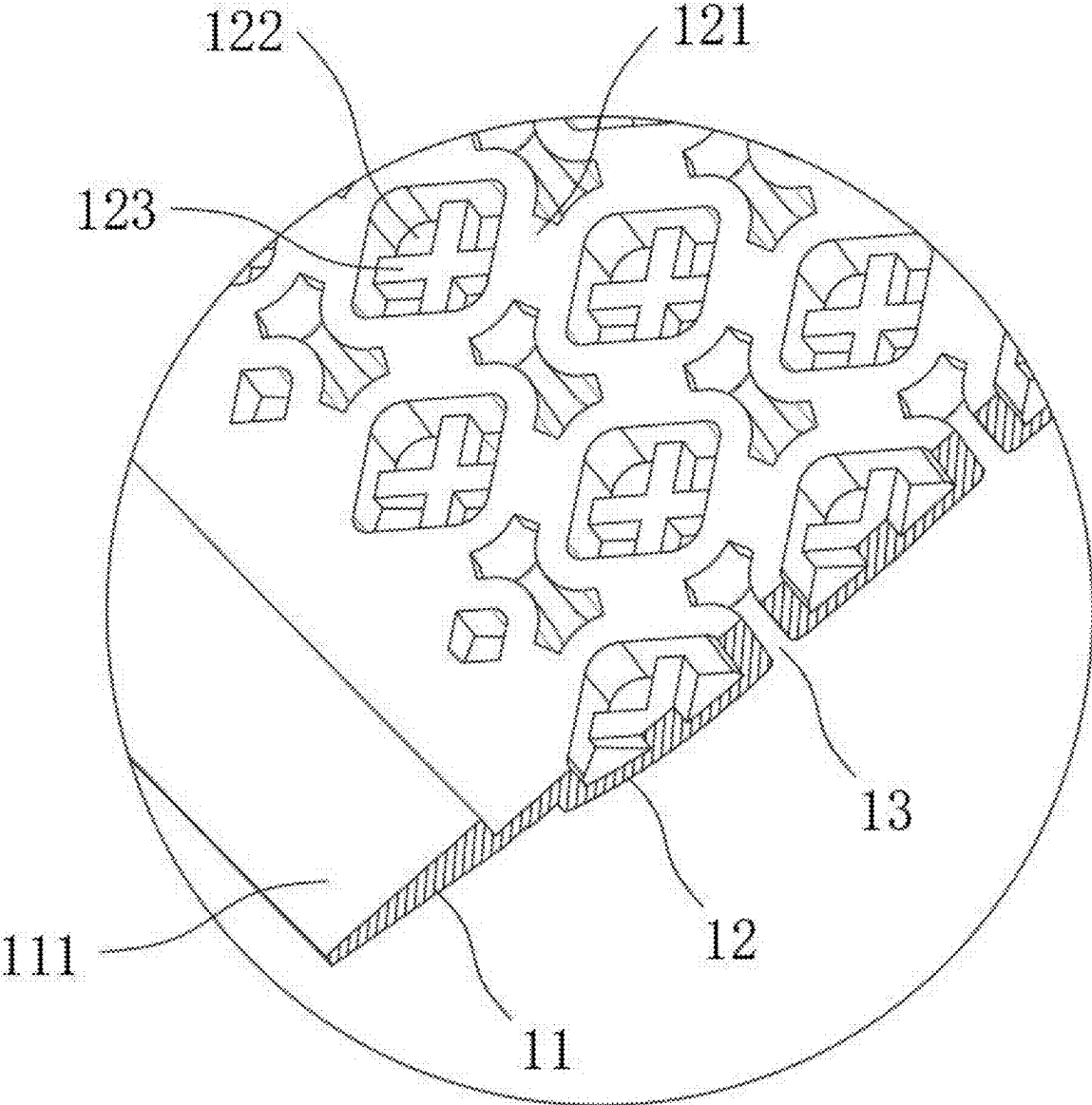


FIG. 3

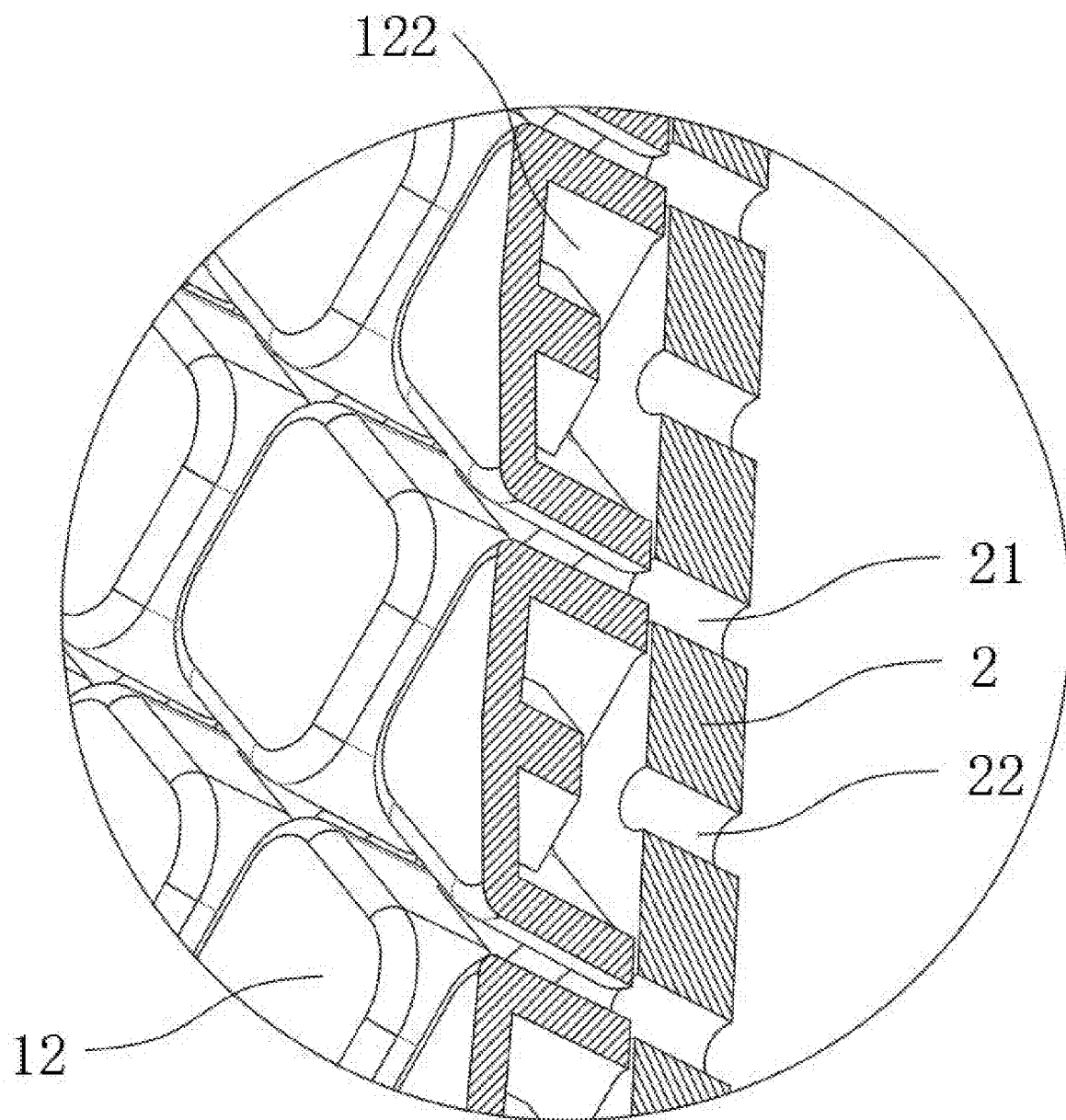


FIG. 4

CUSHIONING STRUCTURE FOR ANTI-COLLISION SPORTS PROTECTOR

TECHNICAL FIELD

[0001] The present disclosure relates to the field of sports protectors, and particularly relates to a cushioning structure for an anti-collision sports protector.

BACKGROUND

[0002] When athletes engage in intense sports such as baseball, hockey, ice skating and rugby, they might inevitably fall or collide. Without protective measures in case of fall or collision, an athlete is easily injured due to direct action of an impact force on his/her body. Therefore, athletes usually wear protectors with protective effects to protect their bodies.

[0003] Most of protectors currently available on the market are provided with hard outer shells, and the outer shells are designed to withstand impact forces caused during collisions. A cushioning layer is arranged inside a hard shell, and the cushioning layer is in contact with an athlete's body, and is capable of minimizing direct impact of the hard shell on his/her body when the hard shell is impacted.

[0004] However, the cushioning layer inside the hard shell serves as an interface for contact with human body. However, the hard shell is in direct contact with the cushioning layer, such that although the hard shell is capable of absorbing part of the impact force, most of the impact force generated when the hard shell collides with the hard object is directly transmitted from the hard shell to the cushioning layer and directly acts on an athlete's body. The impact force absorbed is limited, such that when the athlete collides with the hard object, he/she will still feel pain when the protector is poor in reducing the impact force despite of certain protection by the protector.

SUMMARY

[0005] An objective of the present disclosure is to provide a cushioning structure for an anti-collision sports protector, which further reduces an impact force against the protector, and enhances comfort of using the protector.

[0006] The technical solution used for the cushioning structure for an anti-collision sports protector disclosed by the present disclosure is as follows:

[0007] the cushioning structure includes an energy-absorbing plate and a cushioning plate, and the energy-absorbing plate and the cushioning plate are both made of elastic materials. The energy-absorbing plate includes an annular frame, a plurality of energy-absorbing parts are arranged at an interval inside the annular frame, and two adjacent energy-absorbing parts are fixedly connected through a connecting block. A cavity is formed inside each of the energy-absorbing parts, a plurality of the energy-absorbing parts are combined to form an energy-absorbing net, an inner side of the annular frame is fixedly connected to the energy-absorbing net, and the cushioning plate covers an end of the annular frame.

[0008] Preferably, a reinforcing rib is arranged in the cavity in an extending manner.

[0009] Preferably, a cross section of each of the energy-absorbing parts is polygonal.

[0010] Preferably, an annular step is arranged at an end of the annular frame in an extending manner, and the cushioning plate is snap-fitted into the annular step.

[0011] Preferably, a first ventilation hole is formed between four adjacent energy-absorbing parts, the cushioning plate is provided with a plurality of second ventilation holes arranged at an interval, the second ventilation hole is communicated with the first ventilation hole, an opening is formed at an end of each of the energy-absorbing parts, the cavity is communicated with the opening, the cushioning plate is provided with a plurality of third ventilation holes arranged at an interval, and the third ventilation hole is communicated with the opening.

[0012] Preferably, the energy-absorbing plate is made of an elastic material.

[0013] Preferably, the cushioning plate is made of a foam material.

[0014] The cushioning structure for an anti-collision sports protector disclosed by the present disclosure has the following beneficial effects:

[0015] When the protector is fitted against the athlete's clothing, the cushioning plate is in contact with the athlete's clothing. When a hard object impacts the protector, the hard object first impacts the energy-absorbing parts on the energy-absorbing plate. The cavity allows the energy-absorbing parts to increase their elasticity like an inflatable balloon, such that the energy-absorbing parts are capable of absorbing part of the impact force through the elastic deformation. Moreover, two adjacent energy-absorbing parts are capable of folding through the connecting block therebetween, such that the energy-absorbing plate undergoes an inward recessing deformation process to absorb the impact force. The inward recessing of the energy-absorbing plate compresses the cushioning plate, and the cushioning plate further absorbs the impact force from the hard object, which gradually reduces the impact force from the hard object in a way of using softness to conquer strength, and reduces the athlete's post-impact pain, thereby enhancing the comfort of using the protector.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a schematic structural diagram of a cushioning structure for an anti-collision sports protector according to the present disclosure.

[0017] FIG. 2 is a rear view of an energy-absorbing plate of a cushioning structure for an anti-collision sports protector according to the present disclosure.

[0018] FIG. 3 is a sectional view of an energy-absorbing plate of a cushioning structure for an anti-collision sports protector according to the present disclosure.

[0019] FIG. 4 is a sectional view of an energy-absorbing plate and a cushioning plate of a cushioning structure for an anti-collision sports protector according to the present disclosure.

DETAILED DESCRIPTIONS OF THE EMBODIMENTS

[0020] The present disclosure will be further elaborated and illustrated with reference to specific examples and accompanying drawings in the specification:

[0021] Please refer to FIG. 1.

[0022] In the present disclosure, a cushioning structure for an anti-collision sports protector is disclosed, including an

energy-absorbing plate 1 and a cushioning plate 2, and the energy-absorbing plate 1 and the cushioning plate 2 are both made of elastic materials.

[0023] Please refer to FIGS. 1-3.

[0024] In this example, the energy-absorbing plate 1 is preferably made of an elastic material, and the energy-absorbing plate 1 made of the elastic material has stronger elastic deformability and is lighter, thereby reducing an overall weight of the protector. The energy-absorbing plate 1 includes an annular frame 11, an annular step 111 is arranged on an end face of an end of the annular frame 11 in an extending manner, the annular step 111 is close to an outer side of the annular frame 11, and a mounting groove is formed between an inner side of the annular step 111 and the end face of an end of the annular frame 11.

[0025] A plurality of energy-absorbing parts 12 are arranged at an interval inside the annular frame 11, and an end face of an end of each of the energy-absorbing parts 12 is lower than the annular step 111. A cross section of each of the energy-absorbing parts 12 is polygonal, and in this example, the cross section of each of the energy-absorbing parts 12 is preferably quadrilateral and shaped like a rhombus, such that the energy-absorbing parts 12 of this shape can be arranged more compactly inside the annular frame 11. Two adjacent energy-absorbing parts 12 are fixedly connected through a connecting block 121, a thickness of the connecting block 121 is smaller than that of each of the energy-absorbing parts 12, and the connecting block 121 is located on a straight edge of the corresponding rhombus-shaped quadrilateral energy-absorbing part 12. When two adjacent energy-absorbing parts 12 are impacted, the two adjacent energy-absorbing parts 12 are capable of bending the connecting block 121, thereby creating an effect of recessing in the energy-absorbing plate 1. The energy-absorbing plate 1 absorbs part of an impact force caused by hitting a hard object in the recessing process, such that when the two adjacent energy-absorbing parts 12 are elastically deformed upon impact, sufficient space for deformation is available.

[0026] A cavity 122 is formed inside each of the energy-absorbing parts 12, an opening is formed on the end face of an end of each of the energy-absorbing parts 12, and the cavity 122 is communicated with the opening, such that when the energy-absorbing part 12 is compressed, air inside the cavity 122 can be expelled through the opening. When the energy-absorbing part 12 is impacted, the energy-absorbing part 12 is capable of deforming towards the cavity 122, which enhances a cushioning effect of the energy-absorbing part 12 on the impact force caused by hitting the hard object. Further, the cavity 122 enables the energy-absorbing part 12 to be hollow, thereby further reducing a weight of the energy-absorbing plate 1. A reinforcing rib 123 is arranged at a bottom of the cavity 122 in an extending manner, and in this example, the reinforcing rib 123 is preferably shaped like a cross. Four contact ends of the reinforcing rib 123 are all fixedly connected to an inner wall of the cavity 122, and the contact ends of the reinforcing rib 123 are close to the connecting block 121. When the energy-absorbing part 12 undergoes elastic deformation due to impact by a hard object, the cross-shaped reinforcing rib 123 is capable of enhancing anti-deformability of the energy-absorbing part 12, which further enhances the cushioning effect of the energy-absorbing part 12 on the impact force caused by hitting the hard object. Moreover, when the

energy-absorbing part 12 is no longer compressed by the hard object, the reinforcing rib 123 is capable of pushing the energy-absorbing part 12 back to its original shape, thereby preventing plastic deformation of the energy-absorbing part 12.

[0027] A first ventilation hole 13 is formed between four adjacent energy-absorbing parts 12, and in this example, the first ventilation hole 13 is preferably close to a corner of the energy-absorbing part 12.

[0028] A plurality of the energy-absorbing parts 12 are combined to form an energy-absorbing net, and an inner side of the annular frame 11 is fixedly connected to a plurality of the energy-absorbing parts 12 on an outer periphery of the energy-absorbing net.

[0029] Please refer to FIGS. 1-4.

[0030] In this example, the cushioning plate 2 is preferably made of a foam material, and the cushioning plate 2 made of the foam material has stronger elastic fatigue resistance and better toughness. The cushioning plate 2 is provided with a plurality of second ventilation holes 21 arranged at an interval, the second ventilation holes 21 penetrate through the cushioning plate 2, and in this example, the number of the second ventilation holes 21 is less than that of the first ventilation holes 13. The cushioning plate 2 is provided with a plurality of third ventilation holes 22 arranged at an interval, and the third ventilation holes 22 penetrate through the cushioning plate 2. The cushioning plate 2 covers the end face of an end of the annular frame 11, the cushioning plate 2 is snap-fitted into the mounting groove of the annular step 111, and the mounting groove of the annular step 111 fixes the cushioning plate 2 onto the annular frame 11. The second ventilation hole 21 is communicated with the adjacent first ventilation hole 13, and the third ventilation hole 22 is communicated with the opening.

Usage and Structural Operating Principle

[0031] When the protector is fitted against an athlete's clothing, the cushioning plate 2 is in contact with the athlete's clothing, and the energy-absorbing plate 1 located on an outer side. When an external hard object impacts the protector, the energy-absorbing parts 12 on the energy-absorbing plate 1 are recessed towards the cavity 122 to absorb part of the impact force. Air inside the cavity 122 is compressed and expelled through the corresponding third ventilation hole 22. The cushioning plate 2 is in close contact with the athlete's clothing, and the athlete's skin under the clothing, such that a certain compressive force is required to expel the air from the cavity 122 through the third ventilation hole 22, and the cavity 122 slowly expelling the air enables the corresponding energy-absorbing part 12 to absorb part of the impact force. Moreover, two adjacent energy-absorbing parts 12, by bending the connecting block 121 therebetween, further absorb part of the impact force. At the same time, the two adjacent energy-absorbing parts 12, by bending the connecting block 121 therebetween, compress the cushioning plate 2, and the cushioning plate 2 is elastically recessed to absorb most of the impact force, which gradually reduces the impact force from the hard object in a way of using softness to conquer strength, and reduces the athlete's post-impact pain.

[0032] When an athlete does sports, his/her skin emits heat from the body, and heat emitted by his/her skin covered by the protector is expelled from the clothing, the second

ventilation holes **21** and the first ventilation holes **13** sequentially, which solves the problem of stuffiness in the skin covered by the protector.

[0033] Due to a flat structure of the protector, adjustment to a contour size of the annular frame **11** allows the protector to be placed inside an interlayer of the clothing, such that his/her chest, back, abdomen, arms, legs, knees, and other body parts can be protected, indicating strong expandability. Since both the energy-absorbing plate **1** and the cushioning plate **2** are made of the elastic materials, the protector is capable of bending to fit the athlete's body contour, such that the cushioning plate **2** is more suitable for the athlete's skin.

[0034] The present disclosure provides the cushioning structure for an anti-collision sports protector. When the protector is fitted against the athlete's clothing, the cushioning plate is in contact with the athlete's clothing. When a hard object impacts the protector, the hard object first impacts the energy-absorbing parts on the energy-absorbing plate. The cavity allows the energy-absorbing parts to increase their elasticity like an inflatable balloon, such that the energy-absorbing parts are capable of absorbing part of the impact force through the elastic deformation. Moreover, two adjacent energy-absorbing parts are capable of folding through the connecting block therebetween, such that the energy-absorbing plate undergoes an inward recessing deformation process to absorb the impact force. The inward recessing of the energy-absorbing plate compresses the cushioning plate, and the cushioning plate further absorbs the impact force from the hard object, which gradually reduces the impact force from the hard object in a way of using softness to conquer strength, and reduces the athlete's post-impact pain, thereby enhancing the comfort of using the protector.

[0035] Finally, it should be noted that the above examples are merely intended for description of the technical solutions of the present disclosure rather than limitation to the scope of protection of the present disclosure. Although the present disclosure is described in detail with reference to the above preferred examples, those of ordinary skill in the art should understand that they may still make modifications or equivalent replacements to the technical solutions of the present disclosure without departing from the spirit and scope of the technical solutions of the present disclosure.

What is claimed is:

1. A cushioning structure for an anti-collision sports protector, comprising an energy-absorbing plate and a cushioning plate, wherein the energy-absorbing plate and the cushioning plate are both made of elastic materials;

the energy-absorbing plate comprises an annular frame, a plurality of energy-absorbing parts are arranged at an interval inside the annular frame, and two adjacent energy-absorbing parts are fixedly connected through a connecting block; a cavity is formed inside each of the energy-absorbing parts, a plurality of the energy-absorbing parts are combined to form an energy-absorbing net, and an inner side of the annular frame is fixedly connected to the energy-absorbing net; and

the cushioning plate covers an end of the annular frame.

2. The cushioning structure for an anti-collision sports protector according to claim **1**, wherein a reinforcing rib is arranged in the cavity in an extending manner.

3. The cushioning structure for an anti-collision sports protector according to claim **2**, wherein a cross section of each of the energy-absorbing parts is polygonal.

4. The cushioning structure for an anti-collision sports protector according to claim **3**, wherein an annular step is arranged at an end of the annular frame in an extending manner, and the cushioning plate is snap-fitted into the annular step.

5. The cushioning structure for an anti-collision sports protector according to claim **4**, wherein a first ventilation hole is formed between four adjacent energy-absorbing parts, the cushioning plate is provided with a plurality of second ventilation holes arranged at an interval, the second ventilation hole is communicated with the first ventilation hole, an opening is formed at an end of each of the energy-absorbing parts, the cavity is communicated with the opening, the cushioning plate is provided with a plurality of third ventilation holes arranged at an interval, and the third ventilation hole is communicated with the opening.

6. The cushioning structure for an anti-collision sports protector according to claim **5**, wherein the energy-absorbing plate is made of an elastic material.

7. The cushioning structure for an anti-collision sports protector according to claim **6**, wherein the cushioning plate is preferably made of a foam material.

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